Regulated Research Institutional/Industrial Setting Form (2C)

This form is required for any project where research was conducted at an RRI or any worksite other than home, school or field (one per site). This form must be completed AFTER experimentation by the supervising adult.

Student's Name(s) Aarushi Upadhyayula

Title of Project Additive or Redundant? Combining Gravitational-Wave, sGRB, and Host-Galaxy Constraints on the Distance-Inclination Degeneracy in Binary Neutron-Star Mergers

To be completed by the Supervising Adult in the Setting (NOT the Student(s)) after experimentation:

(Responses must be on the form as it is required to be displayed at student's project booth; please do not print double-sided.)

Research was supported at my work site:

1. The student experience at your work site included:
   - Used equipment and/or received data ☐ Yes ☒ No
   - Minimal interaction with our group ☒ Yes ☐ No
   - Mentored by me or someone else from our group ☒ Yes ☐ No
   - Worked as a sub-set of our ongoing research ☐ Yes ☒ No
   - Had an independent project from our group ☒ Yes ☐ No

2a. Please describe the independent and/or creative work done by the student in any phase of the project, but particularly in developing the hypotheses or engineering goals of the project

The project was the student's own independent work. She formulated the research question - whether short gamma-ray burst and host-galaxy constraints on binary neutron star mergers are additive or redundant in breaking the distance-inclination degeneracy - and developed the hypothesis that both constraints are needed for maximum information gain. She designed the four spatial-geometry test cases and built the complete computational framework herself: simulating gravitational-wave signals with bilby, running dynesty nested sampling, applying Gaussian reweighting to incorporate the electromagnetic constraints, and using KL divergence to quantify the information gained by each pipeline. My role was limited to mentorship: discussing the topic area, giving input on the metrics used, and providing feedback on drafts.

2b. What was the duration of your support of this student project?

Approximately 4 months (late June 2026 to late October 2026).

3. Detail the student's role in conducting the research (e.g. data collection, specific procedures performed). Differentiate what the student observed and the student actually did.

The student personally performed every stage of the research: she wrote the Python code, configured the two LIGO-based interferometer models, simulated the gravitational-wave signals and injected them into Gaussian noise, ran the nested-sampling analyses, produced the posterior corner plots, applied the Gaussian randomization and reweighting for the electromagnetic constraints, and computed and interpreted the KL-divergence results. She also wrote the research paper. I performed no procedures and generated no results; I followed her progress in regular mentoring meetings and provided guidance and feedback on the metrics, writing, and presentation.

4. Did you provide any data; if yes, please provide source or describe

No. I did not provide any data. All data were generated (simulated) by the student herself using the bilby software package; published GW170817 results were used as background literature only.

5. Did the student(s) work on the project as part of a group? ☐ Yes ☒ No
   Were there other high school students present? If yes, please list the students names and describe how their work was related or different from the work of this project.

   No. The student completed this project independently; no other high school students were involved in or present for this project.

6. If this project is under a grant and needs to be acknolwedged, please list the grant statement here.

   Not applicable. This project was not conducted under a grant.

I attest that the student has conducted the work as indicated above and that any required review and approval by institutional regulatory board (IRB/IACUC/IBC) has been obtained. Copies are attached if applicable. I further understand that I will be notified and have access to the public materials to be displayed at Regeneron ISEF (as applicable).

Alyssa Mia Taliotis		Research Mentor
Direct Supervisor's Printed Name	Signature	Title
RISE Global Education		10/28/2026
Institution		Date Signed (must be after experimentation) (mm/dd/yy)
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Education/Experience/Training		Email/Phone